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Graded Assignment • 30 min

Due Mar 20, 11:59 PM CET

1. Place the following events in order from the oldest to the most recent: 1 point

1. the Cretaceous-Paleogene extinction
2. the Permian-Triassic extinction
3. the Cambrian "explosion"
4. the first flowering plants

- (selected) 3,2,4,1
- 1,2,3,4
- 3,2,1,4
- 2,3,4,1

2. What do scientists consider the three main lineages (sometimes referred to as domains) on the Tree of Life? 1 point

- bilateria, radiolara, mesozoa
- Protostome, Deuterostome, Ecdysozoa
- plant kingdom , animal kingdom, microbial kingdom
- (selected) Archaea, Bacteria, Eucarya

3. To calibrate time on the Tree of Life: 1 point

- (selected) A fossil of a known age can be placed on the tree which establishes the minimum age of its sister group and for deeper nodes.
- A fossil of known age can never be reliably placed on a molecular phylogeny and so we can only rely on the molecular data.
- A fossil of a known age can be placed on the Tree of Life to establish the maximum age of nearby nodes.
- All fossils of a known species can be dated and averaged to establish the date of appearance of that lineage.

4. Which of these statements is not true of the movement of life from the oceans onto the land? 1 point

- (selected) The fossil record shows that once an organism has evolved into a land-based organism, its evolutionary diversification then prevents an evolutionary pathway back into marine environments.
- Bacteria and Archaea possibly invaded land-based aqueous systems as long ago as 2.5 billion years ago.
- At least two different invasions of land by arthropods may have occurred.
- The earliest lineages of land plants include mosses, liverworts, and hornworts and appeared on Earth between 400 million to 500 million years ago.

5. As scientists refine the Tree of Life they learn that: 1 point

- The biosphere has been shown to evolve independently of tectonic changes that involve the continents on Earth.
- The independence of the history of life from the geologic processes on Earth has been well established.
- (selected) The Earth's geochemical cycles, climate and atmospheric cycles, and erosion patterns can be linked to the diversity, abundance, and metabolic processes of life at any particular point in time.
- Molecular data is not helpful in establishing the timing of events on the Tree of Life.

6. What does evolutionary devlopmental biology (Evo-Devo) tell us about evolution? 1 point

- (selected) The principal regulatory genes associated with embryonic development are conserved across phyla.
- Each vertebrate species follows a very different development path utilizing its own genetic toolkit to accomplish regulatory processes.
- A single regulatory gene seems to come in play only once during development and this necessitates the large number of genes devoted solely to developmental events.
- Hox genes regulate different developmental tasks in different organisms.

7. As scientists have sequenced proteins, they have found evidence for all of the following EXCEPT which one: 1 point

- All living things are connected, genetically and morphologically, to a single common ancestor.
- The biodiversity that occurs is partly the result of evolutionary changes within regulatory DNA.
- (selected) The expression of regulatory genes at different times and places gives rise to the same structures in different types of organisms creating identical morphology in different creatures.
- The expression of regulatory genes at different times and places gives rise to the diversity seen in different groups of organisms.

8. In researching the lower Congo River: 1 point

- (selected) Scientists have found that beneath the surface of the river, isolated deep canyons and river hydrology has generated barriers that have resulted in allopatric speciation of fish.
- Scientists have found that the large volume of water movement along the river has allowed for a single ecosystem to flourish which contains many unique species found throughout the river system.
- Scientists have tried to map the river system, but present technology limits the ability to measure depths and currents in fast flowing rivers.
- Scientists think that the turbulent hydrology has limited the ability for speciation in this complex environment.

9. Which of the following are NOT factors that may result in background extinctions? 1 point

- environmental changes that result in increased temperature and moisture gradients
- changes in land forms that result in population fragmentation
- (selected) impacts of large asteroids such as the one that struck Earth at the end of the Cretaceous period
- biotic factors such as competition and predation

10. The analysis of biodiversity in Thoreau's woods indicates that: 1 point

- All of the organisms in the woods are thriving due to warmer temperatures.
- All of the organisms in the woods are in decline because of climate change.
- (selected) Organisms that have declined or increased tend to fall in groups of related organisms (clades).
- There has been little change in biodiversity since Thoreau's record keeping.